Bridging the Digital Gap in Global McDonald's Restaurants

D4F Assignment

Use case: Fair Resource Allocation in McDonald’s Edge Computing System

Background
McDonald's operates over 43,000 restaurants worldwide, utilizing AI-driven edge computing to enhance operational efficiency. However, disparities in hardware capacity, network stability, and AI feature availability across locations create fairness challenges. Restaurants in well-connected urban areas benefit from more advanced AI insights, while those in rural or underdeveloped regions may experience slower response times and reduced functionality.

Persona
Jorge, a manager at a McDonald's in a small town in Argentina, relies on AI-driven kitchen automation and predictive maintenance. However, his restaurant’s edge node has lower computational power than locations in major cities. During peak hours, the system struggles to process real-time orders, leading to delays and customer dissatisfaction.

Use case: Fair Resource Allocation in McDonald's Edge Computing System

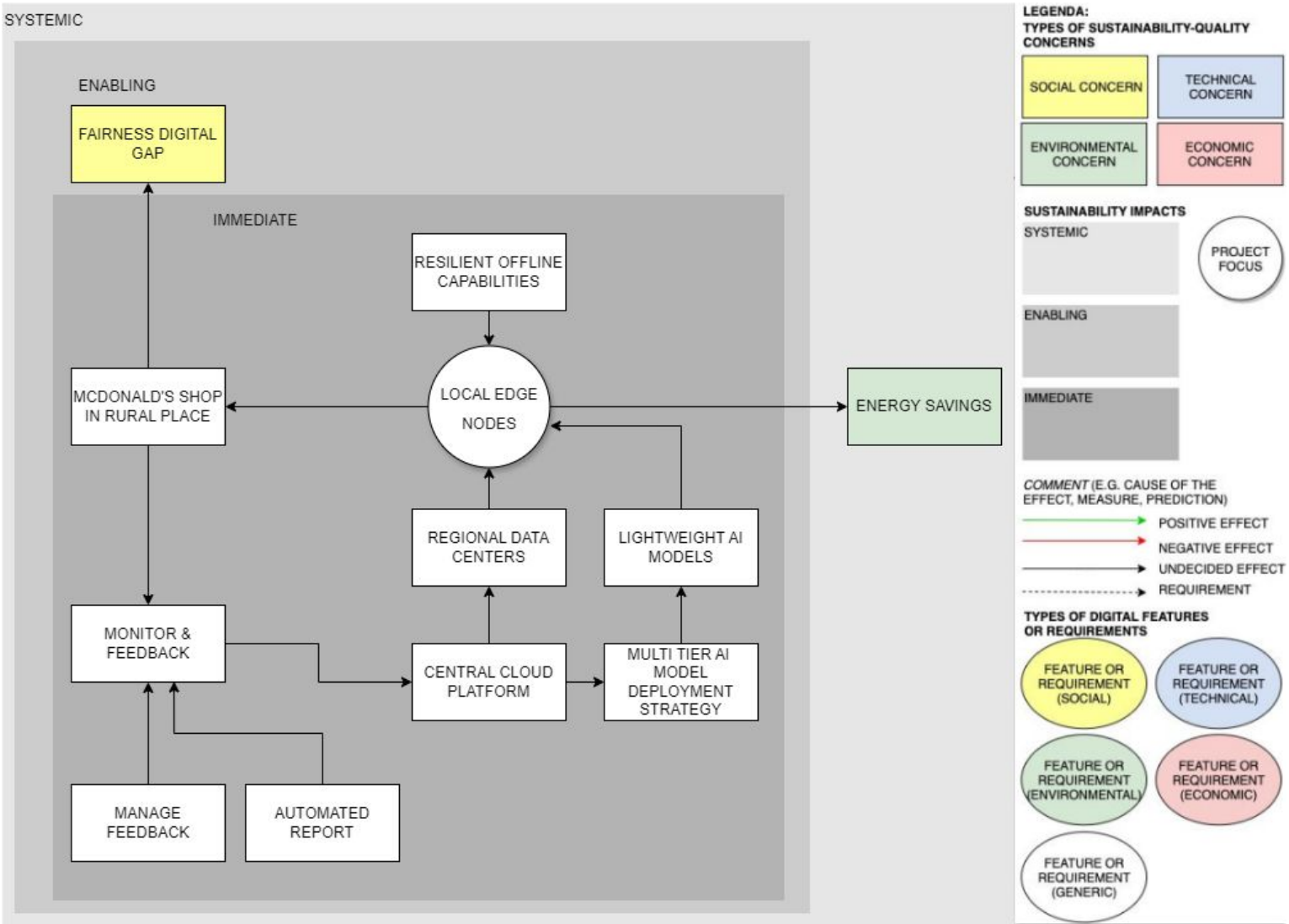
Fairness Challenge

1. Some locations receive priority in AI model updates and processing power.
2. Poor network conditions slow data synchronization.
3. AI-driven order optimization works efficiently in urban stores but is unreliable in rural areas.
4. Fairness expectations and data privacy laws may vary between regions, requiring location-specific adaptations.

Vision

A fairness-oriented edge computing system dynamically redistributes computational workloads based on restaurant needs, ensuring equitable AI performance across all locations. This enhances efficiency, customer satisfaction, and long-term sustainability.

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

The fairness-oriented infrastructure for McDonald's ensures equitable digital access, resilient operations, and sustainability through optimized resource distribution, adaptive AI deployment and continuous feedback integration.

Fairness Digital Gap: Focuses on reducing disparities in digital infrastructure across different McDonald's locations, especially in rural areas.
Lightweight AI Models: Optimized AI models requiring less computational power, making them suitable for low-resource environments(temporary low connection and specific place that cannot get fair computing power even with edge nodes)
Multi-Tier AI Model Deployment Strategy: Hierarchical AI model placement across different infrastructure levels.
Local Edge Nodes: Decentralized computing units that improve processing speed and reduce reliance on central cloud systems.
Monitor & Feedback System: Gathers performance and user feedback to optimize AI decisions and resource distribution.

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	Tech	Eco	Social	Env	DEFINITION of METRIC/KPI
Resource Allocation Coverage	Measures the fairness of AI resource allocation, computing power, and infrastructure across McDonald's restaurants globally.	Resource monitoring system, computation logs					RAF = 1 - (AVG{region}(CPU, Memory, AI Model Access) / SUM({global}(CPU, Memory, AI Model Access))
AI Model Availability	Evaluates how timely AI model updates are accessible across different restaurants, ensuring equitable AI distribution.	AI deployment logs, model versioning system					AMA = (SUM{restaurants} Model_Update_Access) / Total_Restaurants
Lightweight AI Models	Measures the coverage of Lightweight AI Models used across different part of the world, especially for the one in the rural place; the lightweight AI model should be used when there are connection issues	AI deployment logs, model versioning system					AMFA = (SUM{restaurants} Lightweight_AI_Model)) / Total_Restaurants
Latency Consistency	Measures the variance in AI processing time across different restaurant locations to ensure consistent customer experience.	End-to-end transaction logs, customer feedback reports					LC = 1 - (AVG{region}(AI_Response_Time) / SUM{global}(AI_Response_Time))
Resilience in Offline Mode	Evaluates the proportion of AI functions that remain operational during network disconnections to support resource-constrained locations.	Offline resilience module logs, system stability tests					ROM = (AI_Functions_Available_Offline / AI_Functions_Online) × 100%

Definition of Fairness: SQ model (cont.)

The SQ model introduce the QAs of the AI powered infrastructure at McDonald's restaurants.
There are basically two main parts of the QA model: edge allocation and Lightweight AI Model Usage.
The edge allocation helps all the restaurants allocation have the same computing power, which can be test by the key quality attributes of resource allocation fairness
And for the Lightweight AI Model usage, which is used for optimized AI models requiring less computational power, making them suitable for low-resource environments, this can be test by the QA of Lightweight AI Model Coverage.
For the part of latency consistency and resilience in offline mode, both of them can be improved by the result of combined effects of the use of lightweight AI model and edge allocation.

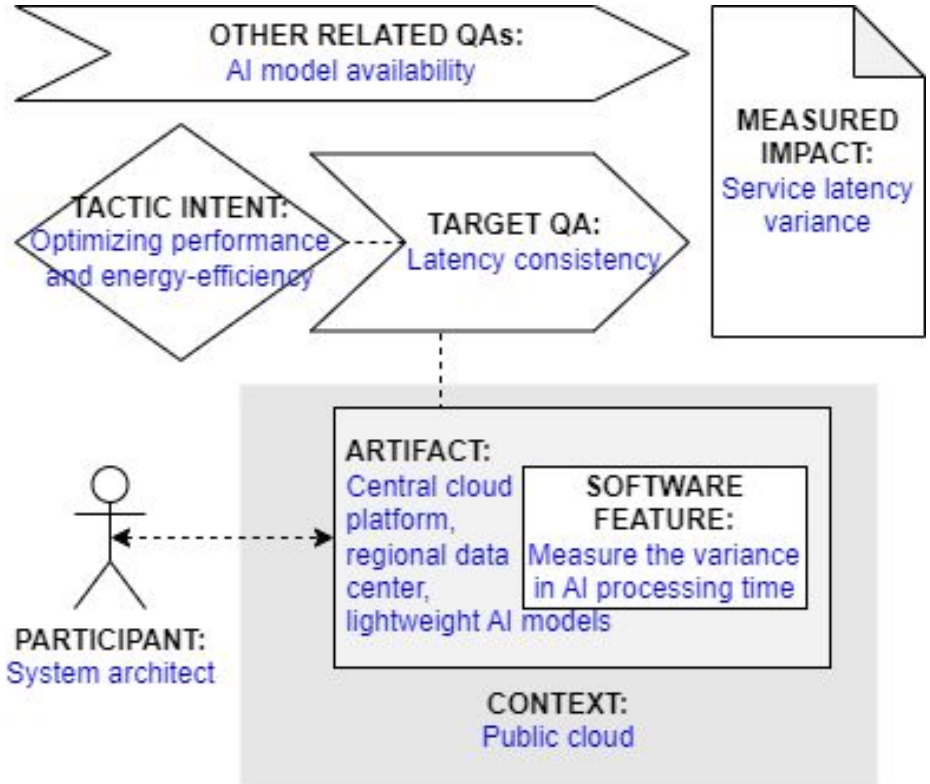
Fairness perspective

Fairness in Edge Computing from a Sustainability Perspective	
DESIRED QUALITY	The goal is to achieve for all McDonald's restaurants, no matter their location or infrastructure, have equal access to AI-driven tools and computing resources
CONCERNS	A system that fairly distributes computing power, customers should experience the same level of AI-driven recommendations and automation, whether in a major city or a remote town, locations with weaker internet connections or outdated hardware need AI models that still work efficiently. Multi-tier AI models based on varies computational power, lightweight AI model are used when there are connection issues
ARCHITECTURAL TACTIC(S)	Dynamically distributes computing power based on real-time demand and regional infrastructure. Deploys lightweight AI solutions that can function even with limited connectivity to gain higher resilience when it is offline also for the places that cannot have enough computiong power even with the design of edge computing
POSSIBLE KPI(S)	Tracks how evenly computing resources are allocated across all restaurants. Measures differences in AI response times between locations to ensure consistent performance. Assesses how much of the system remains functional in low-connectivity scenarios.
PROBLEMS AND PITFALLS	Fairness expectations and data privacy laws may vary between regions, requiring location-specific adaptations.

Definition of KPI*

KPI Name: Computing Resource Fairness Index					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Ensure fair and efficient distribution of computing resources across all McDonald's restaurants to maintain consistent service quality.	***1. Balanced allocation of computing power based on demand and infrastructure.*** ***2. Minimized variance in AI response times across locations.*** ***3. Efficient utilization of edge and cloud computing layers.***	Computing Resource Fairness Score (CRFS) = f(resource availability, demand distribution, variance in allocation)		**Resource Availability =** f(baseline allocation, additional capacity, edge/cloud balance) **Demand Distribution =** f(real-time computational demand, AI processing needs, order volume, connectivity status) **Variance in Allocation =** f(standard deviation of resource allocation across locations)	*–Minimum guaranteed CPU/memory per restaurant* *-Total additional computing resources allocated dynamically per day* *-Average computational load per restaurant (as % of baseline allocation)* *-Number of active AI features per restaurant* *-Standard deviation of allocated resources across all restaurants* *-Percentage of restaurants covered by local edge nodes*
		Target	**Action(s)**		
		- Computing Resource Fairness Score (CRFS) ≥ 0.9 with variance in allocation below 10%.	**Deploy additional local edge nodes more evenly worldwide to enhance computing resource balance.**		

Architectural Tactic(s) for Fairness*



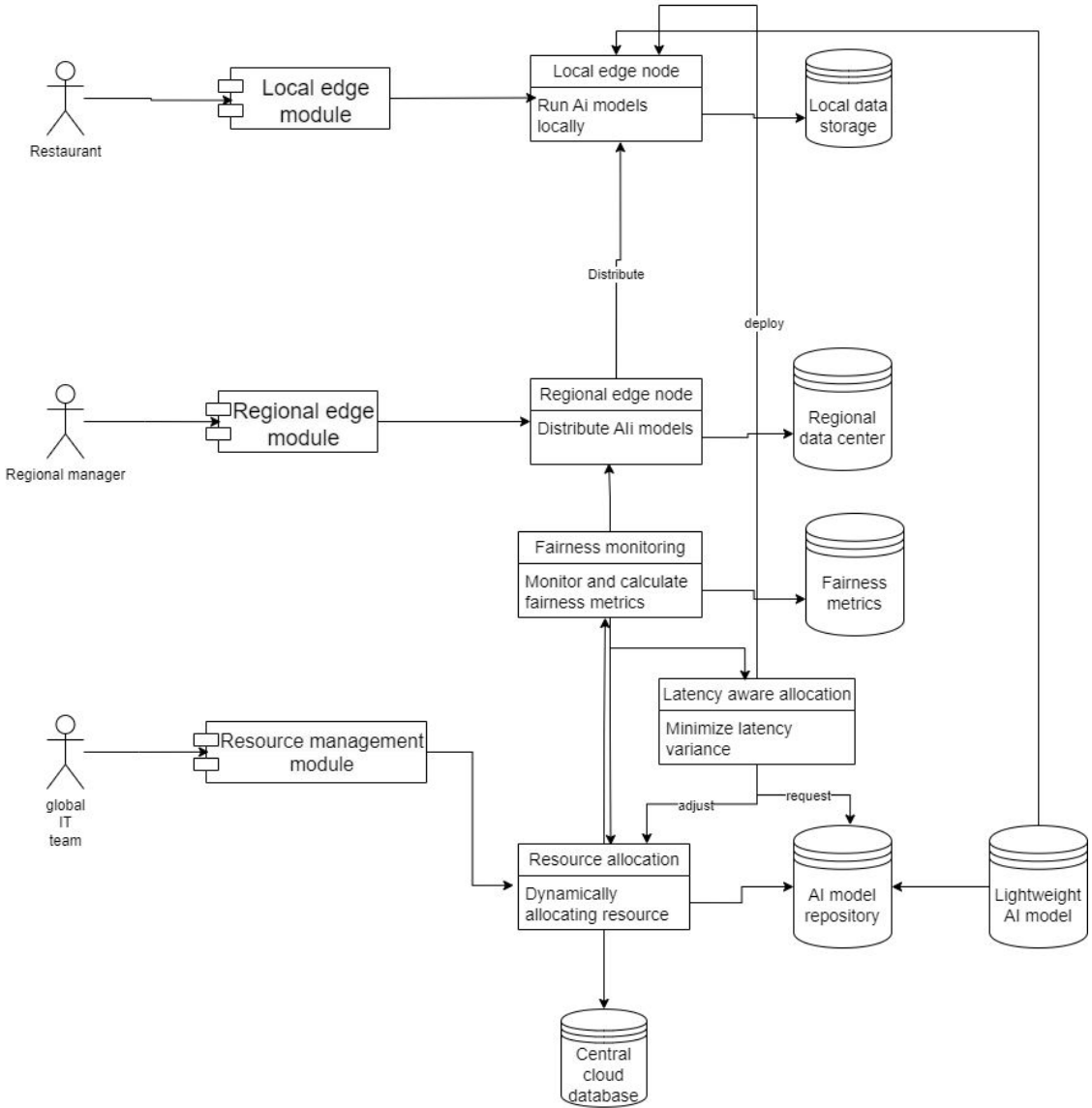
Architectural Tactic(s) for Fairness*

layout: tactic
title: "Latency-aware edge"
tags: Performance
t-sort: "Awesome Tactic"
t-type: "Architectural Tactic"
categories: example-category
t-description: "This tactic aims to minimize latency variance across McDonald's restaurant locations by dynamically allocating computing resources based on real-time demand and infrastructure conditions. It ensures consistent customer experience regardless of location through multi-tier deployment. The tactic addresses performance variation caused by geographic disparity, aiming to support fairness while optimizing response time in AI-driven services."
t-participant: "System architect"
t-artifact: "Central cloud platform, regional data center, lightweight AI models"
t-context: "Public cloud"

Architectural Tactic(s) for Fairness*

t-feature: "Measure the variance in AI processing time"
t-intent: "Optimizing performance and energy-efficiency"
t-targetQA: "Latency consistency"
t-relatedQA: "AI model availability"
t-measuredimpact: "Service latency variance"
t-source: "da Silva Veith A, de Assuncao M D, Lefevre L. Latency-aware strategies for deploying data stream processing applications on large cloud-edge infrastructure[J]. IEEE transactions on cloud computing, 2021."
t-source-doi: "https://doi.org/10.1109/TCC.2021.3097879"

Fairness view: McDonald's Fairness functional view



Fairness view (cont.)

This functional view contains three levels.

Local edge level

Restaurants run AI models locally via edge nodes, supported by lightweight AI models when computational capacity is limited. Data is stored locally to reduce network dependency.

Regional edge level

Regional edge nodes manage model distribution to multiple restaurants.

Global Level

The central cloud platform with a resource allocation module dynamically distributes AI resources. The fairness monitoring module collects real-time performance metrics. Latency aware allocation system adjusts resource distribution to minimize service variance and bring fairness between regions.